

Paper: 1MA1/3H					
Question		Answer	Mark	Mark scheme	Additional guidance
18	(a)	Correct statement	C1	for substituting both 1 and 2 into $x^3 + x$ or into $x^3 + x - 7$	All arithmetic shown must be correct. Ignore any additional trials shown. $x_1 = 1.70997\dots$ $x_2 = 1.74241\dots$ $x_3 = 1.73884\dots$ Accept an accuracy of 2 dp or more rounded or truncated for values of x_1 and x_2 Award the marks for 1.7 on the answer line provided correct iterations are shown in the working space.
			C1	for values 2 and 10 plus explanation that these are above and below 7, or for values -5 and 3 plus explanation that there is a change of sign, thus implying a solution lies between 1 and 2	
	(b)	Correct rearrangement	C1	for correct algebraic rearrangement	
	(c)	1.74	M1	for substitution of 2 into the formula eg $\sqrt[3]{7-2}$ ($= 1.70997\dots$)	
			M1	for a substitution of x_1 to give x_2 ($= 1.74241\dots$)	
			A1	for answer in the range 1.738 to 1.74	